

WHestBench Phase-1 Algorithmic Contribution (short form)

- #326094

Phase-1 Algorithmic Contribution — short-form writeup

Status: SHORT v1, 2026-08-17, prepared for the Aug 17 23:59 UTC filing. This document supersedes the v9 filed 2026-08-10, condenses draft v13 (~8,900 words) to its load-bearing content, and incorporates all thirteen errata recorded against the filed text (§6). Claim levels use the tags of companion P1: **[O]** observed, **[D]** derived, **[R]** reported, **[A]** assumed.

Beyond Gaussian closure: structured spherical quadrature for deep random ReLU networks, and a map of which white-box improvement families fail

o. Summary and provenance

1. We submit a structured spherical-quadrature estimator — a frozen phased-Hadamard exact spherical 2-design, 64,512 directions at the exact chi-mean radius — graded **adjusted $1.832e-7$** (raw final-layer MSE $2.818e-7$) on 50/50 public MLPs with zero failures, 3.5x better than the grader's Monte-Carlo reference. **Graded submission ID: #326094.**
2. The durable contribution is a map of the boundary: proofs and matched measurements of which white-box improvement families cannot work on these networks and why, including an exact two-sided closure of the design axis and an analytic variance prediction accurate to 6.4%.
3. The falsification protocol is the second contribution: twelve predeclared kills, a 267-record fold ledger with negative results retained in full, and adversarial machinery that found and forced the withdrawal of four of our own headline claims during the campaign.

Hosted figures throughout are from the graded ledger for #326094 (a1_hosted_ledger.json): 5.75 s mean / 6.80 s max wall per MLP against the 60 s cap, at C/B 0.650. Locally measured timings are labelled as such and do not transfer across hardware.

Evidence repository — estimator source, predeclarations, kill gates, adversarial audits, frozen manifests, and the 267-record fold ledger:

```
github.com/gmrnk/recursive-estimator-folding
/tree/f225be4e4e4872dc2bef06711525cf00e73a332b/corpus/whestbench
```

Read that path, not the repository root: the default branch is a 2026-08-06 snapshot whose ledger carries 43 records rather than 267. The campaign branch is agent/compression-survivor-corpus; the pinned commit is the pushed branch tip of 2026-08-17, verified against origin before filing, so the citation cannot drift. The repository was made public on 2026-08-17, immediately before this filing, so the citation resolves for any reader. It contains no challenge data, no private truth, no scorer, no credentials, and not the #326094 submission archive.

1. The estimator, and exactly which of its numbers were chosen

The estimator integrates over a frozen phased-Hadamard **exact spherical 2-design**: 126 mutually unbiased frames of 256 directions each, every frame an orthonormal basis $H_{256} \text{diag}(\phi_i)/16$, antipodally doubled to 64,512 points at the exact chi-mean radius, with a per-network Haar rotation as the sole randomization. Five components, each independently ablated on a cached-truth panel with paired seeds and CIs:

1. **The design**, worth 2.02x isolated against radially conditioned Monte Carlo.
2. **Exact radial conditioning**. A bias-free ReLU network is positively one-homogeneous, so $E[f(X)] = E[|X|] \cdot E[f(U)]$ holds exactly at every layer. The radial degree of freedom is not reduced, it is *removed*.
3. **Pilot-rescued structural pruning**. An analytic diagonal pass marks neurons below a threshold as provisionally cold; a 256-antipodal-pair pilot rescues any that fire. Saves **25.109% of B**.
4. **Three-terminal-layer folding**. Dead columns vanish, always-on columns compose linearly into the next weight matrix, and only kink columns retain a ReLU. Saves **4.828% of B** at an MSE ratio of 1.000033.
5. **A first-layer moment-tangent control**, frozen coefficient, measured neutral on this design.

Every value here was read from the deployed method-resolution order (`estimator.Estimator` → `kerdock_v3` → `fold3` → `base_estimator`), not from any single class, and is re-derived mechanically by `scripts/verify_phase1_writeup.py`.

What is forced, with nothing tunable in it. The design itself; the sample count $n_{\text{base}} = 126 \times 256 = 32,256$, which is the design's size and not a budget anyone picked; the exact radius $E[|X|] = 15.98438266660852747\dots$ from the chi moments; and the uniform weights, which are a global minimiser of the quadrature error at every spherical-harmonic degree under a zonal Haar-averaged criterion — though not the unique one, and on the deployed antipodally doubled set not unique at any even degree.

What was selected during development, and is therefore fitted in the sense an auditor cares about. There are seven.

constant	value	role
moment_tangent_lambda	0.9807112198896164	first-layer control coefficient
pilot_base	256	pilot pairs for the pruning rescue
fold_pilot_base	1,024	pilot pairs for the terminal fold
dead_alpha	-2.0	cold-neuron threshold
on_alpha	3.0	always-on threshold in the fold
phase_start	2	first frame of the deployed slice
phase_stop	128	last frame of the deployed slice

All seven are scalar and all were frozen before grading. `on_alpha` deserves its own sentence, because earlier drafts of this paper enumerated **six** and omitted it: it is the mirror of `dead_alpha`, it is live on the deployed path, and it was swept over $\{3.5, 4.0, 5.0\}$ against `dead_alpha` on development data (ledger record 202) with **all arms flat**. That flatness is worth more than the omission cost — it is direct evidence the estimator is insensitive to a dial we were free to choose, which is the property an auditor is actually looking for.

Two further values are frozen but belong in neither column — the backend block height `BLOCK_ROWS = 4,096` and the frame ordering — implementation constants carrying no development selection.

We have stated this wrongly twice. An earlier draft claimed “zero fitted structure anywhere in the estimator.” False. The repair was also wrong, reading values from a base class the deployed subclass overrides. Both were caught within the hour by adversarial audit, and both are left on the page rather than quietly replaced, because a paper arguing for an evidence discipline should show that discipline failing and being caught. The count moved from six to seven for the same reason.

What we claim, precisely: the fitted surface is seven scalars, frozen before grading, confined to budget and correction coefficients, containing nothing that could learn the target, and no component was fit to the evaluation suite. Low measured bias does not prove absence of fitting, and we do not claim it does.

2. Why structured spherical designs plateau, and why completing them does not help

Our estimator integrates over 126 phased-Hadamard frames, each an orthonormal basis $H_{256} \text{diag}(\phi_s)/16$, antipodally doubled to 64,512 points on S^{255} . It is an exact spherical 2-design, and antipodally therefore a 3-design. The natural next question is whether pushing to degree-4 exactness would help. It can be answered exactly, in both directions, and the answer is instructive.

The design cannot reach degree 4 at its current size. [D] The Delsarte–Goethals–Seidel bound for an antipodal spherical 4-design in S^{255} — which is automatically a 5-design, since odd harmonics cancel pairwise — is $2 \cdot C(257, 2) = 65,792$ points. We spend 64,512. We are 1,280 points short, so no reweighting of these nodes under antipodal pair symmetry reaches degree 4.

Exactly one frame count fixes it. [D] For m mutually unbiased bases antipodally doubled, every point sees one inner product at +1, one at -1, 510 at 0, and $512(m-1)$ at $\pm 1/16$, so $\sum_y \langle x, y \rangle^4 = 2 + (m-1)/128$. A 4-design requires $3N/(d(d+2)) = m/43$. Equating and clearing $128 \cdot 43$ gives $10965 = 85m$, hence $m = 129$ **and no other integer** — 130 clears the counting floor and still fails. And $129 = d/2 + 1$ is the maximum number of real mutually unbiased bases in R^d when d is a power of four, which $256 = 4^4$ is. Under the Walsh doubling the complete set has d^2+2d points against a floor of d^2+d , clearing by exactly d at every level: $24/20$, $288/272$, $4224/4160$, $66048/65792$. The completed design is a near-tight antipodal 5-design, over the floor by 0.39%.

And we can say in advance what completing it is worth. [D] For a bias-free He-initialised ReLU network the rotation-averaged two-point function is exactly the iterated arc-cosine kernel $\kappa(c) = (E ||X||^2/d) \cdot \kappa^{32}(c)$, so the estimator’s variance decomposes as $\sum_1 ||f_1||^2 A_1$ against the design defects above. That predicts $v_{126} = 2.4977e-7$ against a measured geomean of $2.6697e-7$ over sixteen fresh networks — **the variance of this estimator is predictable from first principles to 6.4%** — and it puts the degree-4 share of that variance at **0.4497%**.

So completion is worth about half a percent, against a 2.33% break-even set by the point-count cost. [O] Measurement agrees from two further directions: a point-count-matched experiment isolating degree-4 exactness at equal 66,048 points returns 0.176%, CI $[0.970, 1.028]$, $P(\text{better}) = 0.54$, and a committed degree-4 control variate returns +0.42%. Three routes, one predictive, all landing an order of magnitude below the bar.

That agreement is worth more than any of the three alone, because the analytic route explains the other two. Most of the variance is simply not where design strength lives: **86% of it sits at degrees 8 and above**, which no reachable design touches.

And degree 6, where the error actually lives, is unreachable. [D] The measured angular error sits at degree 4 (11% of the iid level) and degree 6 (40%). An antipodal 6-design needs $2^C(258, 3) = 5,658,112$ points, 87.7x our budget; any positive-weight rule needs $\dim P_3(S^{255}) = 2,861,952$, still 44.4x.

The design axis is therefore closed from both sides at once, and not for the reason one would guess. It is not that the design cannot be completed — it can be, exactly, and we did. It is that **completing it perfectly buys nothing measurable**, because the estimator’s residual does not live in low-order design strength. Perfecting a design is not the same as reducing its error, and on these networks the two come apart.

3. What we proved

Each result is stated with its exact scope, and the scopes are narrow on purpose.

Uniform weights are globally optimal, non-strictly (companion P4). [D] Positive-semidefiniteness plus constant row sums make uniform weighting a global minimiser of the quadrature error at every spherical-harmonic degree, and for every nonnegative mixture of degrees, under a zonal Haar-averaged criterion. Strictness is weaker: on the 32,256-point base set uniform is the *unique* minimiser only at degrees 4, 6 and 8, where the kernel is verified positive definite; it is false at degree 2, where the design is exact and every per-frame reweighting is free; and on the antipodally doubled 64,512-point set the estimator actually deploys, uniqueness fails at every even degree — the even-degree kernel has block form $\mathbb{I}_2 \otimes \kappa$, whose kernel contains every antipodally antisymmetric perturbation. A global minimiser is all the closure needs; “the constrained minimiser” would be the wrong phrase and we do not use it. Scope: this closes fixed, output-independent reweighting of a fixed design against a zonal criterion; it does not close adaptive weights, changed point sets, or non-zonal criteria.

The Crofton kink-transect identity is true, and the estimator it induces is dead (companion P2). [O] The Gaussian mean of a bias-free ReLU network equals the Gaussian-weighted surface integral of its gradient jumps over the kink set — verified at machine precision (structural checks to 6.7e-16, two independent transect implementations, 3 predeclared + 20 fresh nets). The unbiased estimator it induces was killed at **176,860x worse variance-per-FLOP than Monte Carlo** on the width-64 depth-8 screen, against a predeclared kill line of 100x. The theorem stands; it is a mathematical result, not an estimator.

A family of truth-free estimators is dead by one identity (companion P6). [D] If an estimator anchors on its own uniform frame mean, the sum-one GLS solution is uniform *identically* — for every ridge and every shrinkage, with no probabilistic model. Four self-anchored GLS descendants that failed in four apparently different ways are one failure, and the identity predicts it without an experiment.

The compute lane is closed: pruning is the only legal rank reduction. [D/O] ReLU commutes only with nonnegative monomial matrices, so the only factorization that passes through the nonlinearity is a per-neuron scaling and permutation — structural pruning. The measurements agree with the algebra: mid-layer exact on-composition returned 0.00% billed reduction (M184: certain-on structure

absent where the network is wide, 2.3x under break-even at depth); the float32 recast found no material f64 lane (M183: 0.0755% of billed predict FLOPs, full recast capped at 59,656,312 FLOPs); the Gen-7 cost-remap attack found the bill ~99% irreducible float32 matmul already at the floor rate. Within the one legal lane, pruning saves 25.109% of B, MSE-neutral (1.014 [0.98, 1.05]) and strictly optimal under the adjusted-score arithmetic, and the terminal fold saves 4.828% of B at an MSE ratio of 1.000033.

The variance is predictable from first principles to 6.4%. [D/O] The iterated arc-cosine kernel decomposition of §2 predicts $v_{126} = 2.4977e-7$ against a measured geomean of $2.6697e-7$ over sixteen fresh networks, puts the degree-4 share of variance at 0.42–0.45% (analytic 0.4497%; committed degree-4 control variate +0.42%), and puts 86% of the variance at degrees 8 and above, where no reachable design has strength. This is what closes the design axis.

What we do not claim: the residual’s harmonic spectrum, any maximum-entropy property, any truncation-class closure, and any minimax floor are all **open** (§6, E4/E5).

4. Negative results, and the method that produced them

The method. Every kill was predeclared with its gate before implementation; none was retuned after seeing results; kills are final and stay in the record. The 267-record fold ledger retains negative results in full rather than pruning them, which is what makes it checkable rather than asserted. The same standard applies to dead entries: a record-level sweep found three historical “kills” that had never actually been run — dispositions carried by wording, not evidence — so we ran 16 falsifiers on the strongest candidates and adjudicated all 16: ten converted from assumption to confirmed measured kills, four revived as screened Phase-2 proposals, one blocked, one held inconclusive.

Twelve predeclared kills. [O]

mutation	measured	verdict
N4 cheap variance levers	null	killed
N5 multilevel closure control variate	1.07x	killed
N6 exact great-circle Rao-Blackwellization	FoM 0.006x	killed
N7 RQMC superconvergence at depth 32	slopes $-0.97/-1.23$ vs -1.25 gate	killed
N8a Kronecker lattice vs our frames	lattice 2.1x worse, CI [1.65, 2.65]	killed
N8b disclosed native backend	$0.94e11$ FLOP/s $< \lambda = 1e11$	killed
N8c offline-trained corrector	bias share -0.0336 vs 0.25 gate (see E6)	killed
N9 frames + tangent + deeper fold	+2.1% (positive control +34.5% on iid)	killed
M180 stronger spherical designs	all arms +20–49% variance	killed
M181 terminal rectified-Gaussian smoothing	bias 4–6x baseline MSE, variance identical	killed

mutation	measured	verdict
M183 float32 hot-path recast	f64 lane 0.0755% of billed predict (E1)	killed
M184 mid-layer exact on-composition	0.00% billed reduction	killed

M180’s kill doubles as a structural result: every perturbation of the frame family or its shared rotation destroys degree-2 exactness and loses 20–49% variance — the design is locally optimal.

The Gaussian-closure wall behind N5/M181. [O] The strongest closure we could certify — exact full-covariance, pairwise-exact via Owen-T/ Φ_2 with per-call enclosure certificates — lands at $9.61e-5$ bias MSE against $\sim 2.5e-7$ for sampling at the same budget. The like-for-like gap is **340.9x raw against the graded raw 2.818e-7** (52.4x adjusted-on-adjusted at the most favourable multiplier; the formerly printed 524x mixed units and is withdrawn, E7). Exact Gaussian structure pays when *subtracted* (the moment-tangent control) and fails when *predicted*. This kills the implementation as an estimator; it does not prove no Gaussian-informed method can work.

The completion kill (MUB129). [D/O] The most attractive open door — completing the 126-frame design to the full 129 real mutually unbiased bases, the maximum in $\mathbb{R}^{256}$ by prior art, yielding an exact near-tight antipodal 5-design — is killed by arithmetic: the analytic decomposition puts the entire degree-4 effect at 0.4497% of variance against a 2.33% point-count break-even, and the point-count-matched measurement (0.176%, CI [0.970, 1.028]) has no power to resolve an effect that small — its band is several times wider than the effect. Three routes, one verdict, an order of magnitude below the bar (§2).

The seed-side rotation null (ledger 242). [O] The one candidate our own theory could not exclude in advance: replace the grader-seed Haar rotation with the deterministic V from the first layer’s SVD, coupling the randomization to the weights we already possess. Measured on the committed public 0–99 net basis under common random numbers, twice-run bit-identical: paired $t = +0.19$ against a $t \geq 3$ kill gate, better on exactly 50 of 100 nets, bootstrap CI on the mean delta $[-3.19e-8, +2.61e-8]$ symmetric about zero, and a drop-top-5 recomputation that flips the sign — a clean null, bought with +3.6e8 billed FLOPs. The mechanism: V is marginally Haar, so coupling to the singular basis buys no alignment with the fixed cubature frame. One null does not close the seed-side lane, and we say so (§6, E10).

Adversarial closure. [O] Twenty adversarial agents mandated to *beat* the champion attacked through six lenses; five returned no candidate, each leaving a named obstruction; seven previously killed families were re-litigated under changed premises and 7/7 held. The one un-probed crack they named — non-smooth cell-membership covariates — was sealed the same day (S18): all 64,512 design directions occupy 64,512 *distinct* first-layer activation cells, so cell identity is a per-point unique label that can memorize a training residual and can never activate on the held-out half; every gated feature set landed below the predeclared fitting-noise cost of $2.63e-5$.

5. LLM involvement

This campaign was conducted end-to-end by large language models operating as agents under human direction: the estimator, the experiments, the ledger, this document and the six companion papers were all written by LLMs — nothing here is human-authored code with LLM assistance; it is the other way round. Because a fluent model produces plausible text whether or not the claim is true, every load-

bearing claim carries an evidence tag ([O]/[D]/[R]/[A], with [GAP] for known holes), and the corpus was audited by two LLM agents from different model families under a sealed evidence charter, plus a twelve-agent refinement pass instructed to refute rather than confirm. That machinery found, in our own flagship claims, a false uniqueness theorem, an unproved exhaustiveness step, a powerless control presented as a control, and an inadmissible measurement supporting a headline — while public scrutiny of this work found no defect; every defect was found by our own machinery pointed at ourselves.

6. Errata and integrity

This document supersedes the v9 filed 2026-08-10 and incorporates all thirteen errata recorded against it; full statements with their arithmetic are in draft v12 §0 and at each marked site. The errata that changed a headline claim:

- **E2** — the 1.65 suite-calibration factor we advised the field to apply was a mean artifact; the median-recomputed panel matches the grader to 0.05%, and the corrected advice is to calibrate on the median.
- **E4** — the central synthesis (residual as maximum-entropy χ^2_1 speckle at the degree-4 boundary) was stated as a theorem, is not one, and is now **open**; its supporting spectral computation is quarantined as inadmissible, and withdrawing it does not revive the older equipartition account, which was never established either.
- **E5** — the S17 floor ratios were computed at $n = 3$ and are retired; at $n = 80$ the distinct-direction ratio is 1.0044, CI [0.8450, 1.1639], so champion and floor are indistinguishable at the resolution we have and no below-floor claim survives.
- **E6** — N8c's printed confidence interval does not contain its own point estimate and is not a usable net-level interval; the bias result stands only as a 3-net, 16-rotation screen observation (−0.0336 against a 0.25 gate), not a zero-bias theorem.
- **E7** — the 524x closure-gap figure mixed raw against adjusted units and is withdrawn; the admissible ratios are 340.9x raw-on-raw and 52.4x adjusted-on-adjusted.
- **E8** — the “exact radial conditioning 2.14x” pillar was a bundled attribution and is withdrawn; the isolated factors are frame design 2.016x and residual radial improvement 1.062x, and “three proven-locally-optimal multiplicative pillars” is withdrawn with it.
- **E10** — “decorrelated at every design pair,” “the ledger certifies PRNG strength,” and “the rotation lane is closed” are each withdrawn as stated above their evidence: the measured correlations are shell aggregates, a finite battery of failed attacks certifies nothing, and one null does not close a lane.

The remaining errata (E1, E3, E9, E11–E13) correct published numbers, provenance labels, and citations without changing any verdict. One matters for this document's own citation: **E12** — the filed citation pointed at the repository's stale default branch; the correct path is the pinned-commit campaign-branch path given in §0.

Word count: 3451